5)

a) $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, B = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$

$A \cdot B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$B \cdot A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

b) $AB = \begin{pmatrix} 8 \end{pmatrix}$, $BA = \begin{pmatrix} 2 & 4 & 6 \\ 0 & 0 & 0 \\ 2 & 4 & 0 \end{pmatrix}$

$A^T B^T = \begin{pmatrix} 2 & 0 & 2 \\ 4 & 0 & 4 \\ 6 & 0 & 6 \end{pmatrix}$, $B^T A^T = \begin{pmatrix} 8 \end{pmatrix}$

c) zu beweisen $\forall_{m,n} A \in \mathbb{R}^{m \times n}$

$\forall_{n,v} B \in \mathbb{R}^{n \times v}$

$\forall m,n,v \in \mathbb{N}: (A \cdot B)^T \overset{?}{=} B^T \cdot A^T$

Seien $A = (a_{ij})_{i,j=1}^{m,n}$

His answer was shit, didn't copy

7) b) $F(\vec{x}) = \begin{pmatrix} x_1 + 2x_2 - 3x_3 \\ 4x_1 + 6x_3 - 5x_2 \end{pmatrix}$

$$F(\vec{x}+\vec{y}) = \begin{pmatrix} (x_1+y_1) + 2(x_2+y_2) - 3(x_3+y_3) \\ 4(x_1+y_1) + 6(x_3+y_3) - 5(x_2+y_2) \end{pmatrix}$$

$$A\vec{x} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & -5 & 6 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

nach BSP 1-12.

c) $A = \begin{pmatrix} a_{11} & a_{1n} \\ a_{m1} & a_{mn} \end{pmatrix}$ $A^T = \begin{pmatrix} a_{11} & a_{m1} \\ a_{1n} & a_{mn} \end{pmatrix}$

1) $T(A)+T(B) = \begin{pmatrix} a_{11} & a_{m1} \\ a_{1n} & a_{mn} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{m1} \\ b_{1n} & b_{mn} \end{pmatrix}$

$$= \begin{pmatrix} a_{11}+b_{11} & a_{m1}+b_{m1} \\ a_{1n}+b_{1n} & a_{mn}+b_{mn} \end{pmatrix}$$

$$= T(A+B) \checkmark$$

2) $\alpha \cdot T(A)$

$T(\alpha A)$

$$\left(\alpha \begin{pmatrix} a_{11} & a_{m} \\ a_{m1} & a_{mn} \end{pmatrix} \right) = \begin{pmatrix} \alpha a_{11} & \alpha a_{1n} \\ \alpha a_{21} & \alpha a_{mn} \end{pmatrix}^{T}$$

$$\alpha \begin{pmatrix} a_{11} & \cdots \\ & \ddots \end{pmatrix}^T = \begin{matrix} \alpha a_{11} & \cdots & \alpha a_{m1} \\ & & \alpha a_{mn} \end{matrix}$$

d) $T(f+g)(x) = T(f)(x)$

$= T(f)(x) + T(g)(x)$

$T(f+g)(x) = f+g\,(3x+4a) = f\,(3x+4a)$
$+ g\,(3x+4a)$

$T(f(x)) + T(g(x))$ ✓

$T(\alpha f(x))$

?

A7)

$F_A : \mathbb{R} \to \mathbb{R}^2$ mit A $F_A(\vec{x}) = A\vec{x}$

linear.

$1 \leq x_1 \leq 2$ $\quad -1 \leq x_2 \leq 1$

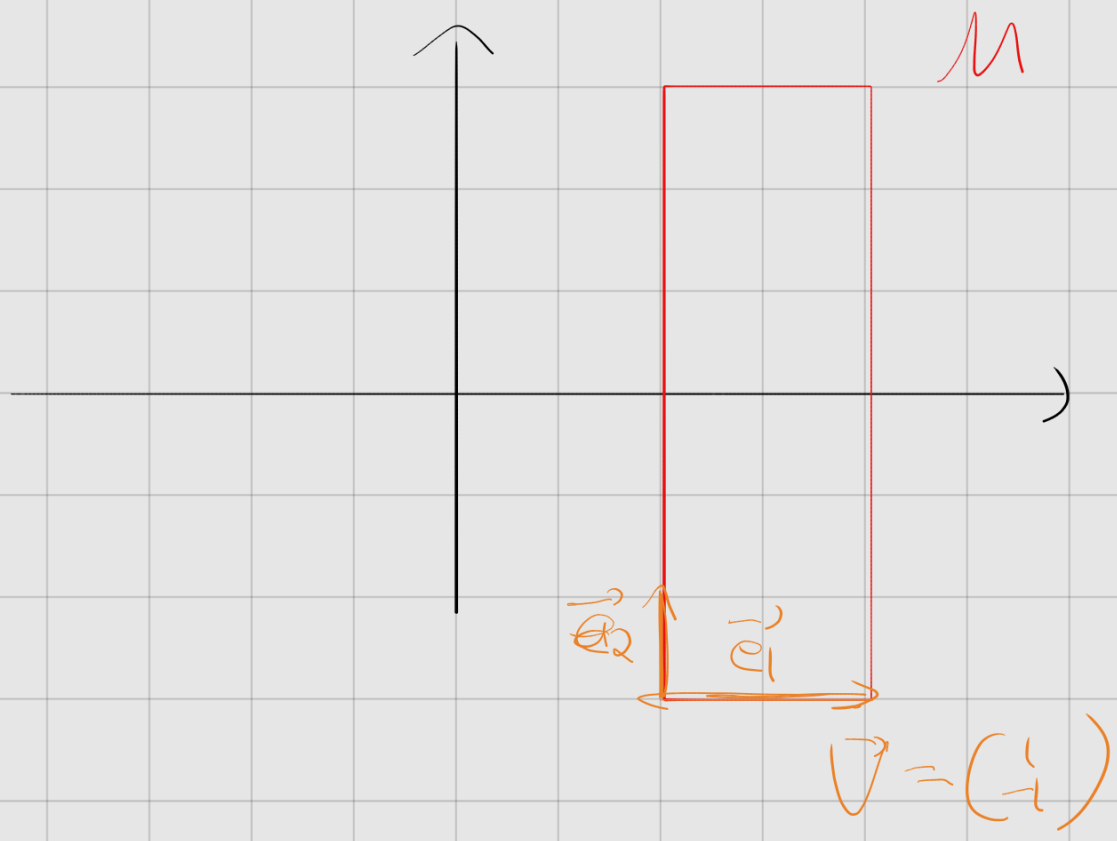
$R_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, $R_2 \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ $R_3 \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, $R_4 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\Rightarrow M$

$A\vec{x}$

$AR_1 \begin{pmatrix} 5 \\ -3 \end{pmatrix}$ $AR_2 \begin{pmatrix} 8 \\ -4 \end{pmatrix}$, $AR_3 \begin{pmatrix} 4 \\ 0 \end{pmatrix}$

$AR_4 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow F_A(M)$



$$M = \{ \vec{v} + \mu e_2 + \lambda \vec{e_1} \mid \mu \in [0,2] \ , \ \lambda \in [0,1] \}$$

$$F_A(\vec{v} + \mu \vec{e_2} + \lambda \vec{e_1}) = F_A(\vec{v}) + F_A(\mu e_2) + F_A(\lambda \vec{e_1})$$